SDU
University of Southern Denmark

Physical contraptions as social interaction catalysts

Mitchell, Robb

Published in:
Proceedings of the 3rd International Workshop on Physicality

Publication date:
2009

Document version:
Proof

Document license:
Unspecified

Citation for pulished version (APA):
Mitchell, R. (2009). Physical contraptions as social interaction catalysts. In D. Ramduny-Ellis, A. Dix, J. Hare, & S. Gill (Eds.), *Proceedings of the 3rd International Workshop on Physicality* (pp. 37-42). Lancaster University. Computing Department.
http://www.physicality.org/Workshops_Archive/Entries/2012/5/4_Proceedings_files/Physicality2009Proceedings(web).pdf

Go to publication entry in University of Southern Denmark's Research Portal

Terms of use
This work is brought to you by the University of Southern Denmark.
Unless otherwise specified it has been shared according to the terms for self-archiving.
If no other license is stated, these terms apply:

- You may download this work for personal use only.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying this open access version

If you believe that this document breaches copyright please contact us providing details and we will investigate your claim.
Please direct all enquiries to puresupport@bib.sdu.dk

Download date: 07. Nov. 2026

Physical contraptions as social interaction catalysts

Robb Mitchell
SPIRE Participatory Innovation Centre
University of Southern Denmark
Alsion 2, Sønderborg 6400 DK
0045 6550 4507
robbathon@gmail.com

ABSTRACT
Can the likelihood of social interactions between strangers be increased by the spatial intervention of interactive physical structures? This paper describes three room-sized mechanical contraptions which were designed to neccesitate the increased physical awareness of, and induce cooperation between, all those people present within a single space. Reactions observed upon the installation of these intended social catalysts at art gallery events are described. A discussion exploring the possible factors contributing to the apparent successes of these contraptions concludes the paper.

Categories and Subject Descriptors
H.5.2 User Interfaces; H.5.3 Group and Organization Interfaces D.2.2 Design Tools and Techniques, J.5 Fine Arts

General Terms
Design, Experimentation, Human Factors

Keywords
social catalyst, installation, intervention, ergomomics, proexemics, design provocations, kinesthetic empathy

1. INTRODUCTION
Two artists, one male, one female stand nude in the narrow doorway of a museum forcing anyone who wished to enter the museum to squeeze in sideways. Visitors could choose for themselves whether they preferred to face the naked female or the naked male. This 1977 performance by Abramowic and Ulay [REF] created for each visitor, a brief, but intense moment in which it was impossible to ignore the presence of the artists.

This example of engineered proximity provided inspiration for the attempts described below to give all visitors to various art gallery spaces an increased physical awareness of other such visitors over more sustained periods whilst allowing everyone to remain fully clothed.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.
Conference'04, Month 1–2, 2004, City, State, Country.
Copyright 2004 ACM 1-58113-000-0/00/0004…$5.00

1.1 Three Social Catalyst Contraptions
Blender, Heads Up of The Table and *Social Whirls* are part of series of art installations by the author designed to foster positive face-to-face interactions between strangers who may not otherwise interact. Each contraption, or to borrow a concept from CSCW research: *social catalyst* [7] presented participants with a shared physical obstacle which was intended to create a situation in which there were no predetermined rules as to how to behave. Providing a novel constraint on "normal" behaviour was intended as a route to dissolve the everyday norms (both internal/individual and social/collective) which may inhibit social interactions. This in turn, could provoke and encourage a fluidity of interaction between strangers.

All the contraptions described were installed for social occasions in art exhibitions in the United Kingdom. The projects are similar in that they all concern face-to-face situations and aimed to provide a mutual experience, context, and a social catalyst for participants, irrespective of and independent of common verbal or written language. All three contraptions also have a wooden finish and their room filling scale creates an obstruction. They all provide the opportunity to dramatically increase the mobility of a normally rigid spatial element of a social environment. Their differences lie principally in which aspect of the environment they each make flexible. *Blender* consists of a move-able walls, *Heads Up of The Table* a moving table top and *Social Whirls* a floor that moves when stepped upon.

1.2 Applications
Although originally developed within a fine art context, this work is also hoped to offer a contribution to a variety of discussions including those around kinesthetic empathy [2] and ethnographic provocation [1] and provide material for reflective design [6] practices concerned with interaction.

2. BLENDER
Is it possible to mix together the dynamism and mobility of an open standing reception with the comfort and focus of a cosy seated gathering?

2.1 Turn dynamics of spatial activity inside-out through constant circulation
Blender (Figure 1) was intended to create a dynamic continuously circulating social situation which might spark interactions between seated and standing guests by providing an intimate but reassuringly temporary space for them to come into contact

Furthermore, it was hoped to invert normal spatial dynamics of a social reception where people on the edge may lean on walls whilst those "working the room" shift around in the centre. An

aim of this contraption was thus to create a social situation in which being a stationary "wall flower" was impossible since the edge of the circular space contained the fastest moving part of the contraption.



Figure 1. plywood, metal and rubber wheeled construction with thirty chairs, 7m x 7m x 2.2m

2.2 "Blender" - a propeller in a seated circle

The contraption developed could equally be likened to a large revolving door or as a giant four bladed non-motorised propeller. This "Blender " was positioned at the centre of a fixed circle of chairs. The four revolving door wooden panels or "blades" were shaped and sized so that they may pass closely over the knees of guests seated on the chairs (Figure 1).



Figure 2. The blades passed close over the knees of seated guests

This revolving door was engineered around a sure axle - the structural column of this ground floor gallery. Thus, despite its considerable weight, it was easy to push the panels from any point except very close to the column (video available). The circle of chairs filled the width of the room and so in order to progress to the rest of the exhibition, refreshments and toilets, guests needed to revolve the barrier by pushing and/or move in the same rotational direction as and when another guest pushed the doors (Figure 2). The blades were constructed of flexible, rounded plywood in order to safely "bounce" off any visitors that were taken unawares by sudden movements.

2.3 "Blender" Observations

As the Blender was rotated the seated and standing guests seemed to find themselves unable not to smile at each other. Seated visitors were frequently observed offering and giving courteous assistance to those standing by trying to push the blades in the direction that those standing were attempting to walk.



Figure 3. Seated viewers could temporarily "see" and "be" in two "rooms" at once

Seated participants commented on appreciating being able to "see round the corner" of the blades or "be in two rooms" at once (Figure 3). When the blade was directly in front of them, they were able to see and so in a sense "be" in two segments at once. This gave them the ability to partially choose their fellow occupants of a segment (both seated and standing) and thus affect who could interact with who. This ability was appeared to be used both very lighthearted (e.g. jokingly "winding people up" by interfering with standing persons interactions with other seated people) and in order to continue interactions once they began. It seemed to be used for choosing to be with fellow segment occupiers of both prior and new acquaintance.

The quantity of interaction between seated participants was unexpected. There were many instances of a full circle of seated participants cooperated in spinning the structure even though they couldn't all see each other. This occurred in two ways: Either some seated participants would join to accelerate the speed of a direction chosen by others. And/or the signal of which way to push was transmitted (by both verbal and nonverbal signals) like along a "chain" by those seated guests who could see each other. On several occasions there were instances of participants developing what could be said to be games with these barriers

including trying to spin the blades as fast as possible and when the structure was moving fast, trying to leave it until the last fraction of second before jumping out it's way. These games occurred as both cooperative play and solo (socially obstructive) exertions.
This energetic play had the effect of encouraging people to sit down.

Walking on seats appeared to be a social taboo that no one was willing to break. However some seated guests shuffled along the seats slowly but even if determined to navigate the obstruction in this way, they eventually found their way blocked by other more steadfastly stationary seated guests.

Several guests that had not visited this particular gallery before were hesitant to approach and touch the contraption irrespective of whether or not it was moving as they did not realise that there was another room and services on the far side of the *Blender*. Many of those whose initial apprehension of the *Blender* was of it steadily rotating in a certain direction, were startled by changes in speed and/or direction of movement. Several made queries and comments regarding what kind of motor was inside it.

The scale of *Blender* meant that it was possible for the blade to come to a halt out of reach of a seated guests who wished to push it. In such instances it was typical that other seated participants that could reach the blades would then "pass" blades to other seated participants in what seemed to be an effort share control, exertion and reward of keeping the contraption moving.

Standing guests improvised a variety of means to negotiate the direction of the room, including calling out through the walls, asking seated guests to pass messages round and looking around the edge of the panels (Figure 4). When people that had simultaneously been standing in different segments met up, they seemed to be connected by their shared experience of the obstruction. Emotions then expressed included apologetic ("sorry that was me pushing that way") and relief "I am glad you weren't in a hurry").



Figure 4. Effecting direction of revolution required negotiation

3. HEADS UP OF THE TABLE

How to reduce guests' tendency to stick close to whoever they arrive with &/or are already familiar with? How to counteract the effect of how as a space becomes smaller, strangers are more likely to appear to actively ignore each other?

3.1 Rotate spatialy prominent roles around a common barrier and hazard

Heads Up Of The Table was developed for a much smaller scale space than *Blender* (Figure) and so the approach was concerned more with aiming to provide a signal unifying talking point and a talking "facility" rather than a division into separate cosy areas.



Figure 5. In situ view: the scale of the revolving table was designed to fill a room

The intention was to invert the typical art gallery spatial configuration where the work displayed often surrounds the viewers and thus visitors are all looking in different directions, and thus not normally looking at each other.

Approaching any social "circle" by oneself may be intimidating and thus is often done discretely, or through the support of a firendly accomplice. This contraption sought to invert such subtle attempts to join a group interacting in space by amplifying the saliency of any entrance into, or exit from the room in which this contraption was installed. A mechanism for isolating and emphasising each entrant to the room was hoped to increase the potential awkwardness of approaching a group to the point of ridiculousness and thus provide a humourous common experience to all gathered around the table.

As the legend of King Arthur makes clear, it is not possible to spatially discren an order of precedence amongst those seated around a round table. Even though the conceived circular table was not perfectly symetrical it was hoped to maintain such equality by rotataing the focus of social attention over time.



Figure 6. Showroom demonstration view of *Heads Up of The Table* to illustrate the whole contraption in action

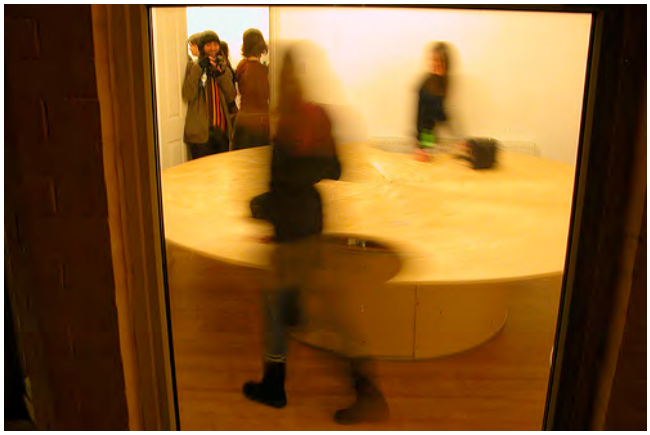


Figure 7. In situ view as seen through external window of room

3.2 "Heads Up" a table to be in and at

A large circular revolve-able table was installed in the centre of an otherwise empty bay-windowed dining room. The diameter of the table surface was only 5cm less than the narrowest point of the (nearly square) room.

Cut away from two opposite ends on the face of the table surface were two circular gaps sized as to allow a person to stand in them. Upon a light push, the table surface rotated (either clockwise or anti-clockwise) so that in order to navigate the room, a participant had to walk into one of the subtracted circular gaps at either opposite "head" of the table and then push the table in the direction of "orbit" that they wished they wished to walk.



Figure 8. Several guests squeezed into the the space designed for a single person

3.3 Observations on "Heads Up of The Table"

Most guests appeared to give the other occupants of the room a larger than normal quantity of attention and the table was responded to with good humour by all who entered it.

Several groups of friends succeeded in squashing two people at a time in the gap designed for one (Figure 8), most of these doubles proceeded to do multiple rotations

The weight of the table top and difficulty of gripping and pushing a thin rounded surface meant that little spinning of table was done by those not positioned within the circular gaps. Thus, contrary to intentions, those at the "heads of the table" experienced the bulk of social attention and physical control of the dominant object within the room. This appeared perhaps to be a heady cocktail for several guests.

Many appeared so transfixed by spinning the table that they directly interacted very little with either friends or strangers in the room. Many guests made prolonged and repeated multiple rotations of the room. This meant that others wishing to enter or leave the room during their circuits had to ask for the assistance of those continually pushing. It appeared that whilst in the room, guests did not discuss much other than the table.

4. SOCIAL WHIRLS

How to turn the socially unproductive nervous energy of standing receptions into something less rigid and more touchy feely?

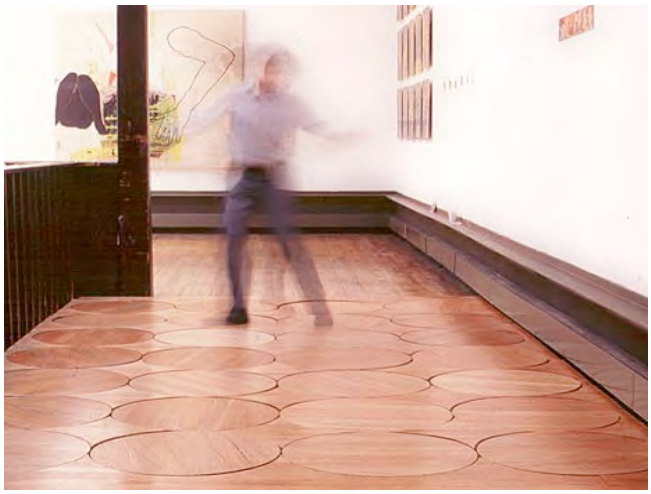


Figure 9. Social Whirls comprised 28 independently revolve-able circular floor panels

4.1 Loosen inhibitions through movement and fear

It was hoped that Social Whirls (Figure 9) would provide a catalyst for spontaneous for physical contact between people for reasons of safety and/or reassurance for example hand-holding and shoulder steadying is both more likely and socially acceptable in situations where the floor is unstable.

It has been said of standing social receptions that people rotate in a slow, but awkward dance. The provision of a dynamic floor was hoped to question and accelerate such movements to the point where awkwardness was dissovled.



Figure 10. Visitors were perhaps more attentive to the floor than each other.

4.2 Floor panels that spun with body weight

Revolve-able circular floor panels were installed to fill a balcony that connected two halves of large museum space. This parquet style floor was designed to match the appearance of the floorboards of the existing museum. Inset into this floor were 28 circular discs. Each of these discs were mounted on unseen ball-bearings which individually rotated when stepped upon.



Figure 11. Many visitors danced despite the absence of music

4.3 "Social Whirls" observations

Social Whirls appeared well received with much smiling and laughter from most guests probably due to a variety of causes such as embarrassment at being unsteady on their feet or a light hearted giddiness. Although there was much hand holding observed this was appeared to only between people with existing bonds. Few instances of physical contact between strangers was observed.

Guests differed very widely in terms of their frequency, duration and intensity of interaction with this contraption. Most brief visitors were understandably nervous and in concentrating their gaze downwards to the floor (Figure 10) were giving the strangers around them no more attention than usual. Many prolonged/repeated visitors invented various games to play on the contraption. For instance, despite the absence of music the contraption inspired much enthusiastic dancing, particularly in a "twist" style (Figure 11). Many strangers dancing at the same time were observed exclaiming to each other both verbally and through physical gestures. Children viewed the floor as a play surface suitable for sitting on and manipulating with their hands (Figure 12).



Figure 12. Children also interacted with the floor whilst seated

5. DISCUSSION

5.1 Limitations of these trials

Many of the concepts of the contraptions appeared successful and that, furthermore additional means of fulfilling their broad aims were discovered during testing. However these apparent successes should be qualified by acknowledging the limited context in which they were installed and a consideration of other factors which may have contributed to their acting as social catalysts.

A principal limitation, is that the audiences at an art gallery event are likely to be more accepting and welcoming of novel experiences than people in most other social situations. Ascertaining if such contraptions have potential for improving social interactions in other contexts requires further investigation. Also it maybe instructive to conduct similar trials in different countries where the norms of social interaction and spaces within which it occurs may be very different. For instance, in contrast to the western concept of space as empty areas or room, to the Japansese space is a living uninterrupted flow always rich in complex interactions between people and objects [5]

Additionally in these art venue trials only very initial short term responses were recorded. Social relationships commonly develop over much longer periods of time and if people that first me through or within a contraption went on to have further encounters, it is not know if these were positive or negative.

A longer period of participants' exposure to the contraptions would also have proved instructive. It is possible that the sheer novelty of the objects was a major factor in enhancing the social atmosphere around the contraptions. One could speculate that when elevators were a novelty, travel between floors was more of a social occasion than now when they are commonplace.

5.2 A playful puzzle

Although the *Blender* seemed to spark the most interaction between strangers, the responses described suggest that all of these projects succeeded to varying degrees as social catalysts within these particular contexts in both expected and unexpected ways. What is not clear is what were the most important factors in this success. Did these projects appear to produce interactions through successfully innovating concepts in spatial intervention or was the successes due more to other formal or contextual factors?

Given that many participants found pleasurable ways to interact with the objects either when they were solo participants and/or ways to manipulate the objects without being particularly interactful with other participants it seems apt to consider whether the most important factor in participant reaction was seeing the objects as some kind of toy and/or the situation as a game. It seems reasonable to speculate that when adults willingly engage in almost any group play (particularly non competitive games) that a lighthearted atmosphere which promotes the likelihood of social interactions will result.

None of these projects were labeled as toys. It was through a process of discovery and exploration that participants discovered, shared, discussed and tested such playful properties. As a result of discovering for themselves such potential, players perhaps felt a sense of ownership of their games and hence were keen to play on for such long periods.

Although the moving parts of these contraptions were relatively simple, the fact that the ball bearings and castors were hidden from view (Figure 13) created for some viewers an initial sense of

confusion and apprehension. The use of transparent materials to produce contraptions in which all working parts were clearly visible could result in less "trial-and-error activity" [7] by participants in their initial exploration of the contraptions and a clearer representation of the "task state" at all times.

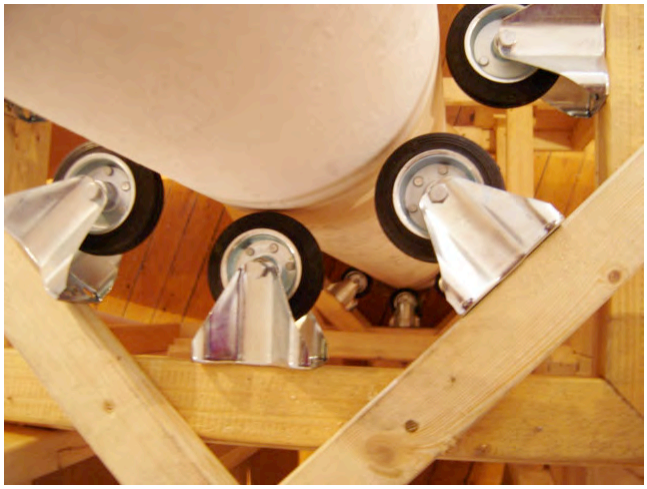


Figure 13. The non visible internal workings of *Blender* as seen from above

5.3 Other contributory factors

All of the contraptions were of a wooden either a large number and/or large scale of curves. It is likely that positive reactions to the properties of wood could have contributed to overall success of the projects. For example, many commented on the nice smell of sawdust and/or would stroke the curves of the wood. Such signaling of a simple easily non controversial pleasure may have contributed to a sense of solidarity amongst participants.

In rooms adjacent to where all three contraptions were installed were more conventional art gallery spaces within which visitors could socialise and view non-interactive artworks. It is possible that the contraptions appeared successful as they acted like a kind of filter or magnet attracting venue visitors most interested in physical activity and spontaneous interactions to focus their gallery visit around playing in and around the contraptions. If so, then it seems reasonable to assume that visitors more inclined to gentler and/or more sedentary viewing experiences would gather elsewhere in the venues - away from the contraption.

5.4 Small group space as small group catalyst

Although mindful of the limitations and other potential contributory factors outliined above, the following tentative comparision is proposed: Appropriately scaled semi-enclosed space seems the most important attribute that a contraption may provide.

Social Whirls and the table contraption both provided a visually open space whereas the *Blender* was much more enclosing. The floor based work could perhaps be more accurately described as a series of individual contraptions closely spaced together. Thus in contrast to the other two pieces, Social Whirls allowed for more independent exploration by separate individuals, although given the simplicity of the contraption, the individual responses by Social Whirls visitors varied less. The openness of *Heads Up of The Table* and the greater ease with which it might be monopolised by one or two users seem to reduce its effectiveness as a social catalyst. The cosy spaces temporarily provided by the panels of the *Blender* appeared more conducive to social interaction than the more unified and all encompassing experience environment engendered by the table and the common activities facilitated by Social Whirls.

However, the range of unpredicted responses to the these relatively simple mechanicsms, and other accidental discoveries outlines the value of further questioning of assumptions concerning the relationship between space and interaction through

5.5 Further work

The audiences at an art gallery event are more accepting and welcoming of novel experiences than most other social situations. If such contraptions have potential for improving social interactions in other situations requires further investigation.

6. ACKNOWLEDGMENTS

Thanks to Simon Harlow, Chris Boyd, Ben Radcliffe, Hannah Clinch and other members of The Chateau and Glasgow School of Art for technical consultation, assistance, equipment and encouragement. These contraptions were produced also with support from Glasgow International Festival of Contemporary Arts and Arts Council England. Informative discussions with Ellen Christiansen, Jared Donovan, Portia File and Ben Matthews are also gratefully acknowledged.

7. REFERENCES

[1] Buur, J. and Sitorus, L. Ethnography as Design Provocation. *Ethnographic Praxis in Industry Conference Proceedings*, (2007), 146–157.

[2] Fogtmann, M.H. Kinesthetic Empathy Interaction–Exploring the Possibilities of Psychomotor Abilities in Interaction Design. *Proceedings of the Second International Workshop on Physicality*.

[3] Karahalios, K. and Donath, J. Telemurals: linking remote spaces with social catalysts. *Proceedings of the SIGCHI conference on Human factors in computing systems*, (2004), 615–622.

[4] Michalak, K. Performing Life, Living Art: Abramovic/Ulay and KwieKulik. *Afterimage 27*, 3 (1999).

[5] Random, M. *Japan*. Crucible, 1987.

[6] Sengers, P., Boehner, K., and David, S. Reflective design. *Proceedings of the 4th decennial conference on Critical computing: between sense and sensibility*, (2005), 49–58.

[7] Sharlin, E., Watson, B., Kitamura, Y., Kishino, F., and Itoh, Y. On tangible user interfaces, humans and spatiality. *Personal and Ubiquitous Computing 8*, 5 (2004), 338–346.